

Zain Khan

kn.zain@hotmail.com | linkedin.com/in/zain-khan | zainkhan.dev |  github.com/7kzaincode

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science (BASc), Computer Engineering

Sep. 2025 – Apr. 2030 (Expected)

– Awarded over **\$40,000+** in merit-based scholarships including the **University of Waterloo Presidential Scholarship**.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C/C++, Java, C#, HTML/CSS, SQL

Frameworks: React, Next.js, Node.js, Express.js, FastAPI, TensorFlow, OpenCV, Three.js, Tailwind CSS

Databases & Infra: PostgreSQL, MongoDB, Supabase, Firebase, Docker, Linux, Vercel, Git

Concepts: System Design, API Design, ML Pipelines, RAG, DSA, OOP, Concurrency, Auth, Caching

EXPERIENCE

WATonomous

Waterloo, ON

Robotics Software Engineer — EVE Autonomy

Jan. 2026 – Present

- Developed **concurrent autonomous robotics software in C++ (ROS 2 Humble)**, coordinating **5+ real-time nodes** across perception, planning, and control subsystems on an autonomous delivery platform.
- Designed a **modular ROS 2 node architecture** spanning perception, world modeling, planning, and control layers, enabling isolated unit testing and deterministic replay in Gazebo simulation environments.
- Implemented **costmaps, A* path planning, and Pure Pursuit control** to achieve stable autonomous point-to-point navigation in cluttered indoor environments with dynamic obstacle avoidance.

Kasperia

Remote

Software Developer (Frontend & Data) — Contract

Jun. 2023 – Sep. 2025

- Improved page load performance by **30%** and reduced maintenance overhead by **25%** by architecting reusable React components with modular CSS across the entire client-facing web application.
- Built **automated data pipelines** with JavaScript, SQL, and REST APIs to consolidate reporting workflows, cutting manual reporting time by **40%** and eliminating recurring data entry errors across teams.
- Developed interactive KPI dashboards with **React and Chart.js**, boosting stakeholder engagement by **35%** through real-time visibility into key business performance metrics and trends.

PROJECTS

otto  | Next.js, TypeScript, Python, LiveKit WebRTC, Deepgram, Gemini 2.5 Flash, Supabase

- Built an **AI voice agent** automating **GitHub, Gmail, and Google Calendar** workflows through natural speech commands; awarded **1st Place — LiveKit Track** at the NexHacks hackathon.
- Implemented a **LiveKit WebRTC pipeline** with **Deepgram VAD** turn detection and **Gemini 2.5 Flash Realtime** native audio inference, achieving sub-second end-to-end voice-to-action response latency.
- Integrated **OAuth 2.0 auto-refresh** via **Supabase Auth** for secure, uninterrupted voice commands across GitHub, Gmail, and Google Calendar services without requiring manual re-authentication.
- Compressed contextual payloads with **TTC Bear-1**, reducing token usage from **4000 → 1800 (55%)** for lower inference latency and significantly reduced API cost per session.

toothfairy  | Next.js, TypeScript, FastAPI, Python, TensorFlow, OpenCV, Gemini 2.5 Flash, Moorcheh AI

- Built a **dental X-ray segmentation pipeline** using a **U-Net (TensorFlow/Keras)** model with **OpenCV Connected Component Analysis** and morphological post-processing, isolating **28+ individual teeth** from panoramic dental radiographs for per-tooth pathology detection and clinical analysis.
- Integrated **Moorcheh AI** for **per-patient RAG** with isolated memory namespaces and semantic search across visits, enabling longitudinal retrieval for clinical history and cross-visit diagnostic correlation.
- Orchestrated **3 AI copilots** (imaging, notes, treatment) using **Gemini 2.5 Flash** for structured JSON extraction and **Gemini Vision** for radiological pathology detection with confidence scoring across **10 conditions**.

Sketchy  | Next.js, TypeScript, Gemini 2.5 Flash, tldraw, MongoDB, GitHub OAuth, Vercel API

- Built a sketch-to-site web app that converts hand-drawn wireframes into fully deployable HTML/CSS/JS via AI-powered code generation, achieving full wireframe-to-production conversion in **under 60 seconds**.
- Optimized a **Gemini 2.5 Flash inference pipeline** with constraint-driven prompting, reducing token usage from **3000 → 800 (73%)** while maintaining output fidelity across diverse wireframe inputs.
- Implemented a **CI-style deployment pipeline** via GitHub and Vercel APIs to auto-generate repos, push source, and return live production URLs, with **GitHub OAuth** and **MongoDB Atlas** persistence.